Volume and Surface Area in 3D

GeoGebra 3D Calculator · GeoGebra

Years 9-10	**~30 minutes**	**Mathematics · Measurement (3D Geometry)**

NOTICE — AI-generated worksheet
This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://www.geogebra.org/3d
Curriculum links: AC9M9M01 · AC9M10M01

Learning intentions

- I can use GeoGebra 3D to build prisms, cylinders and composite solids and identify their dimensions.
- I can calculate the volume and surface area of right prisms, cylinders and composite objects.
- I can explain how changing one dimension of a solid affects its volume and surface area.

Getting started

Open GeoGebra 3D Calculator (geogebra.org/3d). Use the toolbar to construct a rectangular prism first — look for the 'Extrude' or solid-construction tool, or build one from a base polygon. Rotate the view by dragging with your mouse so you can see all faces of the solid before you start recording measurements.

Tasks

1. Construct a rectangular prism with length 4, width 3 and height 5 units. Record its dimensions and calculate its volume by hand, then check your answer using GeoGebra's measurement tools.

2. Using the same prism, calculate its total surface area by hand. Which faces are congruent to each other?

3. Build a cylinder and test at least four different radius values while keeping height fixed. Record the radius and resulting volume each time.

Radius (units)	Height (units)	Volume (units^3)

4. A cylinder has radius 3 units and height 8 units. Calculate its volume and curved surface area by hand (use $\pi \approx 3.14$), then build the cylinder in GeoGebra to check your working.

Working:

5. Combine a rectangular prism and a triangular prism (or a prism and a cylinder) to build one composite solid. Explain the strategy you used to find its total volume.

6. Calculate the volume of your composite solid from the previous task by finding the volume of each part and adding them together. Show your working.

Working:

7. Rotate your composite solid so you are looking directly along one axis. Sketch what you see and describe why this 2D view doesn't show the object's full shape.

8. A cylindrical water tank needs to hold at least 5000 litres (5 m^3). Use GeoGebra to test at least two different radius/height combinations that would work, and explain which one you'd recommend and why.

Extension (optional)

Design a composite solid made from at least three shapes that has a volume between 200 and 250 cubic units. Record the exact dimensions you used and show the calculation that proves your solid meets this target.